

SAT UZBEKISTAN

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Module-1

In triangle ABC and triangle DEF , sides AB and DE each have a side length of 10 inches, and angles A and D each have an angle measure of 40° . Which of the following additional pieces of information is(are) sufficient to prove whether triangle ABC is congruent to triangle DEF ?

- I. The measures of angles B and C are equal.
- II. The lengths of sides AC and DF are equal.
- III. The lengths of sides BC and EF are equal.

- (A) I is sufficient, but II and III are not.
- (B) II is sufficient, but I and III are not.
- (C) III is sufficient, but I and II are not.
- (D) II is sufficient and III is sufficient, but I is not.

2.

The functions f and g are defined by the equations shown, where a and b are integer constants, $a < b$ and $b < 0$. If $y = f(x)$ and $y = g(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function?

- I. $f(x) = a(4.2)^{x+b}$
- II. $g(x) = a(4.2)^x + b$

- (A) I only
- (B) II only
- (C) I and II
- (D) Neither I nor II

3.

A triangular prism has a height of 9 centimeters (cm) and a volume of 234 cm^3 . What is the area, in cm^2 , of the base of the prism? (The volume of a triangular prism is equal to Bh , where B is the area of the base and h is the height of the prism.)



4.

The function f is defined by $f(x) = -39^x$. The function g is a decreasing linear function. In the xy -plane, the graphs of $y = f(x)$ and $y = g(x)$ intersect at two points, (h, j) and (k, m) , where $j > m$. When $g(x) < f(x)$, which of the following must also be true?

☐ (A) $x > k$

☐ (B) $x < h$

☐ (C) $x > k$ or $x < h$

☐ (D) $h < x < k$

5.

$$P(t) = 260(1.03)^{\left(\frac{3}{2}\right)t}$$

The function P models the population, in thousands, of a certain city t years after 2009. According to the model, the population is predicted to increase by $n\%$ every 8 months. What is the value of n ?

☐ (A) 0.22

☐ (B) 1.03

☐ (C) 2

☐ (D) 3

6.

$$x(kx - 96) = -8$$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k ?

7.

A right triangular pyramid has exactly six edges. Each of these edges is 96 centimeters long. If the surface area of the pyramid is $k\sqrt{3}$ square centimeters, what is the value of k ?

8.

Triangle ABC is similar to triangle DEF , where A corresponds to D and C corresponds to F . Angles C and F are right angles. If $\tan(A) = \sqrt{3}$ and $DF = 53$, what is the length of $\overline{DE}$?

☐ A) $53\frac{\sqrt{3}}{3}$

☐ B) $53\frac{\sqrt{2}}{2}$

☐ C) $53\sqrt{3}$

☐ D) 106

9.

At different times on a certain day, a manager at a theme park recorded the number of visitors. An exponential model gives the estimated number of visitors, V , at the theme park t hours after the theme park opened, where $t \leq 3$. Based on the model, 80 visitors were estimated to be at the theme park when it opened, and every 30 minutes the estimated number of visitors increased by 150% of the estimated number of visitors 30 minutes earlier. Which of the following equations best represents this model?

- A) $V = 80(2.50)^{2t}$
- B) $V = 80(2.50)^{\frac{t}{30}}$
- C) $V = 80(1.50)^{2t}$
- D) $V = 80(1.50)^{\frac{t}{30}}$

10.

For data set A, the table summarizes the distribution of the number of pieces of mail received by a business each day during a period of 11 days.

Pieces of mail	Days
0	2
4	2
5	2
6	2
7	1
8	1
17	1

The data value 17 is removed from data set A to create data set B, which consists of the remaining 10 data values. Which statement best compares the median of data set A and the median of data set B?

- A) The median of data set B is less than the median of data set A.
- B) The median of data set B is greater than the median of data set A.
- C) The median of data set B is equal to the median of data set A.

11.

At a hotel, guests used the pool for a total of t hours in March. At the same hotel, guests used the pool for $3.61t$ hours in April. What is the percent increase in the number of hours guests used the pool from March to April?

A) 2.61%

B) 3.61%

C) 261%

D) 361%

12.

In the xy -plane, the point $(6, 7)$ is a solution to which of the following systems of inequalities?

A) $x < 3$
 $y > 5$

B) $x < 3$
 $y < 5$

C) $x > 3$
 $y < 5$

D) $x > 3$
 $y > 5$

13.

$$\begin{aligned} 2x + 5y &= 9 \\ 12x + 30y &= 54 \end{aligned}$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

A) $(r, -\frac{2r}{5} + \frac{9}{5})$

B) $(-\frac{2r}{5} + \frac{9}{5}, r)$

C) $(-\frac{2r}{5} + 9, \frac{2r}{5} + 54)$

D) $(\frac{r}{6} + 9, -\frac{r}{6} + 54)$

14.

$$9x^2 + 30x + q = 43$$

In the given equation, q is a constant. One solution to the given equation is $\frac{-5+\sqrt{14}}{3}$. What is the value of q ?

A) 54

B) 43

C) 11

D) 5

15.

x	y
-6	-14
-3	-16
3	-20
6	-22

The table shows four values of x and their corresponding values of y . The linear relationship between x and y can be represented by the equation $2x + By = C$, where B and C are constants. What is the value of C ?

16.

$$f(x) = -3x^2 + 30x - 48$$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the maximum value of the function as a constant or coefficient?

A) $f(x) = -3x^2 + 24x + 6x - 48$

B) $f(x) = -3x(x - 8) + 6(x - 8)$

C) $f(x) = -3(x - 5)^2 + 27$

D) $f(x) = -3(x - 8)(x - 2)$

17.

An object's speed, in meters per hour, is increasing at a rate of 1,530 meters per hour per second. Which of the following is closest to this rate in **feet per second squared**? (Use 1 meter = 3.28 feet.)

A) 0.13

B) 1.39

C) 7.77

D) 83.6

18.

$$m(t) = -0.0274\left(\frac{t}{7}\right)^2 + 7.3873\left(\frac{t}{7}\right) + 75.032$$

The function m gives the predicted body mass $m(t)$, in kilograms (kg), of a male giraffe t days after it was born in a wildlife reserve, where $t \leq 390$. Which of the following is the best interpretation of the statement " $m(210)$ is approximately equal to 272" in this context?

A) The predicted body mass of a male giraffe was approximately 210 kg 272 days after it was born.

B) The predicted body mass of a male giraffe was approximately 272 kg 210 days after it was born.

C) The predicted body mass of a male giraffe was approximately $\frac{210}{7}$ kg 272 days after it was born.

D) The predicted body mass of a male giraffe was approximately 272 kg $\frac{210}{7}$ days after it was born.

19.

$$\frac{5}{9}(x - k) = k(x - \frac{9}{5})$$

In the given equation, k is a constant. The equation has no solution. What is the value of k ?

A) $-\frac{5}{9}$

B) $-\frac{9}{5}$

C) $\frac{9}{5}$

D) $\frac{5}{9}$

20.

The kinetic energy K , in joules, of an object is given by the formula $K = \frac{1}{2}mv^2$, where m is the mass of the object, in kilograms, and v is the velocity of the object, in meters per second. If the kinetic energy of a certain object can be found by using the formula $K = 82v^2$, what is the mass of the object, in kilograms?

A) 6,724

B) 164

C) 82

D) 41

21.

Parallelogram $PQRS$ is similar to parallelogram $TUVW$, such that P , Q , R , and S correspond to T , U , V , and W , respectively. If $PQ = 28TU$, then which of the following must be true?

A) $PR = 3QR$ B) $PR = 3VW$ C) $QR = 28TU$ D) $QR = 28UV$

22.

A business owner purchased an energy-efficient air conditioning system for a cost of \$4,200, including installation. The business owner saves \$940 per year to operate the new system compared to the operating cost of the old system. Including the cost of the new system, which function f gives the total amount, in dollars, the business owner has saved x years after purchasing the new system?

A) $f(x) = -4,200x + 940$ B) $f(x) = -940x - 4,200$ C) $f(x) = 940x - 4,200$ D) $f(x) = 4,200x - 940$

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Module-2

1.

x	$g(x)$
-17	-7
-9	0
-1	7

For the linear function g , the table shows three values of x and their corresponding values of $g(x)$. Function g is defined by $g(x) = ax + b$, where a and b are constants. What is the value of $a + b$?

2.

32 teams are competing in a baseball tournament. The tournament has 5 rounds. After each round, half the teams competing are eliminated from the tournament. If r teams remain in the tournament after x rounds of play, which of the following gives r in terms of x ?

$$r = \frac{1}{2}(32^x)$$

$$r = \frac{1}{2}x^{32}$$

$$r = 32\left(\frac{1}{2}\right)^x$$

$$r = 32x^{\frac{1}{2}}$$

3.

A boat traveled a certain distance upriver and then traveled downriver 10 miles farther than it traveled upriver. The boat took x hours to travel upriver at a speed of 12 miles per hour. The boat then took y hours to travel downriver at a speed of 18 miles per hour faster than the speed at which the boat traveled upriver. Which of the following equations best represents this situation?

$$12x - 30y = -10$$

$$12x + 30y = -10$$

$$12x + 30y = 10$$

$$12x - 30y = 10$$

4.

A circle has center O , and points R and S lie on the circle. In triangle ORS , the measure of $\angle ROS$ is 116° . What is the measure of $\angle RSO$, in degrees? (Disregard the degree symbol when entering your answer.)

5.

In 1980, the population density of Carroll County was 125 people per square mile. In 1990, the population density of Carroll County was 141 people per square mile. The population density of Carroll County in 1990 was $p\%$ of the population density of Carroll County in 1980. What is the value of p ?

116.0

112.8

16.0

12.8

6.

A cube has an edge length of 13 inches. What is the volume, in cubic inches, of the cube?

39

78

169

2,197

7.

Data set A: 8, 11, 12, 15, 21, 27

Data set B: 9, 12, 13, 16, 22, h

Data sets A and B each consist of 6 values as shown, where h is a constant. If the standard deviation of data set A is greater than the standard deviation of data set B, which of the following could be the value of h ?

- I. 23
- II. 28
- III. 29

(A) I only

(B) II only

(C) III only

(D) II or III

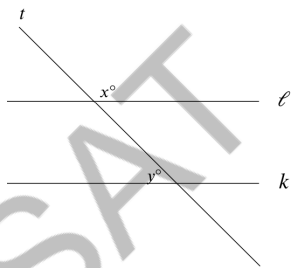
8.

Rectangles $ABCD$ and $EFGH$ are similar. The length of each side of $EFGH$ is 6 times the length of the corresponding side of $ABCD$. The area of $ABCD$ is 96 square units. What is the area, in square units, of $EFGH$?

9.

The expression $(9x - 28)(15x + 4)$ is equivalent to the expression $ax^2 + bx + c$, where a , b , and c are constants. What is the value of b ?

10.



Note: Figure not drawn to scale.

In the figure shown, lines ℓ and k are parallel and line t intersects both lines. If $x > 145$, which of the following must be true?

$y < 35$

$y > 35$

$x + y < 180$

$x + y > 180$

11.

The ratio of cups of flour to cups of butter in a pie dough recipe is 2.75 to 1. The ratio of cups of flour to cups of butter in a biscuit recipe is 7 to 1. Each recipe has 1 cup of butter. How many more cups of flour are in the biscuit recipe than in the pie dough recipe?

12.

An event planner is planning an ice-skating event. It costs the event planner a onetime fee of \$45 to rent the skating rink and \$10.75 per attendee. The event planner has a budget of \$275. What is the greatest number of attendees possible without exceeding the budget?

13.

In triangle ABC and triangle DEF , sides AB and DE each have a side length of 6 inches, and angles A and D each have an angle measure of 40° . Which of the following additional pieces of information is(are) sufficient to prove whether triangle ABC is congruent to triangle DEF ?

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14.

The cost of renting a piece of equipment is \$64*n* for the first day and \$32*n* for each additional day, where *n* is a positive integer. Which of the following functions gives the cost *C*(*x*), in dollars, of renting this equipment for *x* days, where *x* is a positive integer?

(A) $C(x) = 32nx + 32n$

(B) $C(x) = 32nx + 64n$

(C) $C(x) = 64nx + 32n$

(D) $C(x) = 64nx - 32n$

15.

$$31(x - n) = 31y + 31n$$

One of the equations in a system of two linear equations is given, where *n* is a positive constant. The system has no solution. Which equation could be the second equation in this system?

(A) $3x - 3y = 6n$

(B) $3x + 3y = 3n$

(C) $3x + 3y = 6n$

(D) $3x - 3y = 3n$

16.

Which of the following expressions has a factor of $x + 2b$, where *b* is a positive integer constant?

(A) $3x^2 + 9x + 18b$

(B) $3x^2 + 24x + 18b$

(C) $3x^2 + 30x + 18b$

(D) $3x^2 + 39x + 18b$

17.

The composition of an animal is defined as the muscles, bones, and fat of the animal. A scientist studied the composition of one young swamp buffalo and determined the buffalo had 133 kilograms of muscle, which made up approximately 65.9% of its composition. Of the remaining composition of this buffalo, approximately 46.2% was bone, and the remainder was fat. Based on these approximations, to the nearest tenth, how many kilograms of this buffalo's composition was bone?

18.

In right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin(R)$ is $\frac{3\sqrt{17}}{13}$. What is the value of $\cos(S)$?

(A) $\frac{4\sqrt{17}}{51}$

(B) $\frac{3\sqrt{17}}{13}$

(C) $\frac{13\sqrt{17}}{51}$

(D) $\frac{3\sqrt{17}}{4}$

19.

In triangle XYZ , angle Y is a right angle, the measure of angle Z is 39° , and the length of $\overline{YZ}$ is 22 units. If the area, in square units, of triangle XYZ can be represented by the expression $k \tan 39^\circ$, where k is a constant, what is the value of k ?

20.

At a depth of h meters below the surface, the estimated total pressure P_1 , in kilopascals (kPa), a diver experienced is given by $P_1 = 10h + 101$, which is a combination of estimated water pressure and estimated atmospheric pressure. After maintaining a certain depth, the diver descended d additional meters. The estimated total pressure P_2 , in kPa, the diver experienced after descending these d additional meters is given by $P_2 = 10d + 311$. Which of the following is the best interpretation of 311 in this context?

(A) The estimated total pressure, in kPa, the diver experienced at a depth of d meters

(B) The estimated increase in the total pressure, in kPa, the diver experienced for each increase in the depth by 1 meter

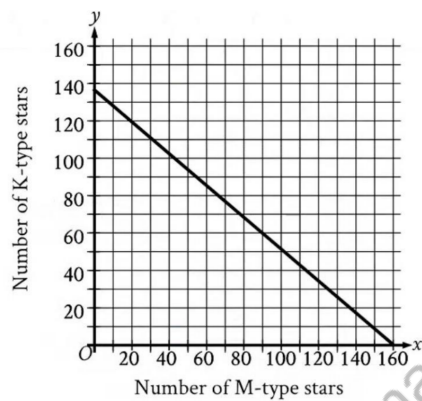
(C) The estimated increase in the total pressure, in kPa, the diver experienced for each increase in the depth by d meters

(D) The estimated total pressure, in kPa, the diver experienced when the diver began to descend d additional meters

21.

The function f is defined by $f(x) = (x - 9)(x - 6)(x - 5)$. In the xy -plane, the graph of $y = g(x)$ is the result of translating the graph of $y = f(x)$ up 4 units. What is the value of $g(0)$?

22.



A certain open star cluster contains M-type stars and K-type stars. The estimated total mass of M-type and K-type stars in this open star cluster is 132,778 quettagrams. The graph shown models the possible combinations of the number of M-type stars, x , and K-type stars, y , that could be in this open star cluster if all the M-type stars have the same estimated mass and all the K-type stars have the same estimated mass. Based on the graph, which of the following is closest to the estimated mass, in quettagrams, of each M-type star in this cluster?

(A) 828

(B) 973

(C) 52,992

(D) 79,786